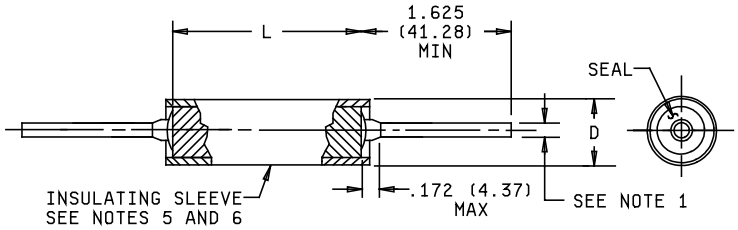
INCH-POUND
MIL-PRF-39022/1G
24 October 2011
SUPERSEDING
MIL-PRF-39022/1F
30 June 2006

PERFORMANCE SPECIFICATION SHEET

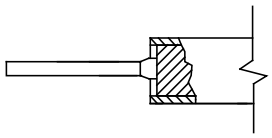
# CAPACITORS, FIXED, METALLIZED, PAPER-PLASTIC FILM, OR PLASTIC FILM DIELECTRIC, DIRECT CURRENT, (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CHR09 (INSULATED)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

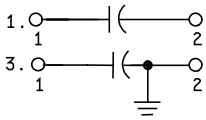
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39022.




CASE FOR CIRCUIT 1



CASE FOR CIRCUIT 3




CIRCUIT DIAGRAM

NOTES:

1. Number 24 AWG wire .020 ± .002 inch (0.51 ± 0.05 mm) for case diameters of .175 inch (4.45 mm) and .195 inch (4.95 mm).
   Number 22 AWG wire .025 ± .002 inch (0.64 ± 0.05 mm) for case diameters of .235 inch (5.97 mm) and .312 inch (7.92 mm).
   Number 20 AWG wire .032 ± .002 inch (0.81 ± 0.05 mm) for case diameters of .400 inch (10.16 mm) and over.
2. See table I for additional dimensions.
3. Dimensions are in inches.
4. Metric equivalents are given for general information only.
5. Insulating sleeve shall extend beyond the capacitor body but shall not exceed .031 inch (0.79 mm) on either end. Insulating sleeve thickness shall not exceed .016 inch (0.41 mm).
6. Plastic insulating sleeve shall be transparent; marking shall be applied to the capacitor case.
7. Metric equivalents are in parentheses.
8. Lead length may be a minimum of one inch (25.4 mm) long for use in tape and reel packaging when specified in the ordering data.

FIGURE 1. Dimensions and configuration.



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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case material: Nonmagnetic (end seals may be of magnetic material).

Capacitance value: See table I

Capacitance tolerance: See table I.

Rated voltage: See table I.

Dielectric material:

50 volt units normally polyethylene-terephthalate.

200, 400, and 600 volt units normally paper polyethylene-terephthalate.

Operating temperature range:

50 volt units: -55°C to +85°C.

200, 400, and 600 volts units: -55°C to +125°C.

Failure rate level (%/1000 hours): M (1.0), P (0.1), R (0.01), and S (0.001).

Radiographic inspection (X-ray): Method 209 of MIL-STD-202.

Thermal shock: Method 107 of MIL-STD-202, condition A except step 3 shall be at applicable high test temperature.

Seal: Method 112 of MIL-STD-202, condition B, or (for liquid-filled units only) 1 hour at applicable high test temperature with no evidence of leakage.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202.

Terminal to terminal: 200 percent of rated voltage.

Terminals to case (when case is not a terminal): 200 percent of rated voltage.

For 100 percent inspection:

250 percent of rated voltage for not less than 5 seconds, or

200 percent of rated voltage for not less than 15 seconds.

Insulation resistance (IR): Method 302 of MIL-STD-202.

Terminal to terminal: See table II.

Terminal to case (when case is not a terminal): 10,000 megohms, minimum.

Capacitance: Method 305 of MIL-STD-202.

Dissipation factor (DF): 1 percent.



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TABLE I. Electrical characteristics, dimensions, and dash numbers.

| DC rated voltage (Volts) | Capacitance value (μF) | Circuit diagram | Capacitance tolerance (Percent ±) | Dimension, nominal (inches) 1/ 2/ | | Dash number Failure rate level (%/1,000 hr) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | L ± .062 | D ± .031 | M(1.0) | P(0.1) | R(0.01) | S(0.001) |
| 50 | 0.018 | 1 | 5 | .750 | .175 | 1261 | 1521 | 1781 | 2041 |
| 50 | 0.018 | 1 | 10 | .750 | .175 | 1262 | 1522 | 1782 | 2042 |
| 50 | 0.018 | 3 | 5 | .688 | .175 | 1263 | 1523 | 1783 | 2043 |
| 50 | 0.018 | 3 | 10 | .688 | .175 | 1264 | 1524 | 1784 | 2044 |
| 50 | 0.022 | 1 | 5 | .750 | .175 | 1265 | 1525 | 1785 | 2045 |
| 50 | 0.022 | 1 | 10 | .750 | .175 | 1266 | 1526 | 1786 | 2046 |
| 50 | 0.022 | 3 | 5 | .688 | .175 | 1267 | 1527 | 1787 | 2047 |
| 50 | 0.022 | 3 | 10 | .688 | .175 | 1268 | 1528 | 1788 | 2048 |
| 50 | 0.027 | 1 | 5 | .750 | .175 | 1269 | 1529 | 1789 | 2049 |
| 50 | 0.027 | 1 | 10 | .750 | .175 | 1270 | 1530 | 1790 | 2050 |
| 50 | 0.027 | 3 | 5 | .688 | .175 | 1271 | 1531 | 1791 | 2051 |
| 50 | 0.027 | 3 | 10 | .688 | .175 | 1272 | 1532 | 1792 | 2052 |
| 50 | 0.033 | 1 | 5 | .750 | .175 | 1273 | 1533 | 1793 | 2053 |
| 50 | 0.033 | 1 | 10 | .750 | .175 | 1274 | 1534 | 1794 | 2054 |
| 50 | 0.033 | 3 | 5 | .688 | .175 | 1275 | 1535 | 1795 | 2055 |
| 50 | 0.033 | 3 | 10 | .688 | .175 | 1276 | 1536 | 1796 | 2056 |
| 50 | 0.039 | 1 | 5 | .750 | .195 | 1277 | 1537 | 1797 | 2057 |
| 50 | 0.039 | 1 | 10 | .750 | .195 | 1278 | 1538 | 1798 | 2058 |
| 50 | 0.039 | 3 | 5 | .688 | .195 | 1279 | 1539 | 1799 | 2059 |
| 50 | 0.039 | 3 | 10 | .688 | .195 | 1280 | 1540 | 1800 | 2060 |
| 50 | 0.047 | 1 | 5 | .750 | .195 | 1281 | 1541 | 1801 | 2061 |
| 50 | 0.047 | 1 | 10 | .750 | .195 | 1282 | 1542 | 1802 | 2062 |
| 50 | 0.047 | 3 | 5 | .688 | .195 | 1283 | 1543 | 1803 | 2063 |
| 50 | 0.047 | 3 | 10 | .688 | .195 | 1284 | 1544 | 1804 | 2064 |
| 50 | 0.056 | 1 | 5 | .750 | .235 | 1285 | 1545 | 1805 | 2065 |
| 50 | 0.056 | 1 | 10 | .750 | .235 | 1286 | 1546 | 1806 | 2066 |
| 50 | 0.056 | 3 | 5 | .688 | .235 | 1287 | 1547 | 1807 | 2067 |
| 50 | 0.056 | 3 | 10 | .688 | .235 | 1288 | 1548 | 1808 | 2068 |
| 50 | 0.068 | 1 | 5 | .750 | .235 | 1289 | 1549 | 1809 | 2069 |
| 50 | 0.068 | 1 | 10 | .750 | .235 | 1290 | 1550 | 1810 | 2070 |
| 50 | 0.068 | 3 | 5 | .688 | .235 | 1291 | 1551 | 1811 | 2071 |
| 50 | 0.068 | 3 | 10 | .688 | .235 | 1292 | 1552 | 1812 | 2072 |
| 50 | 0.082 | 1 | 5 | .750 | .235 | 1293 | 1553 | 1813 | 2073 |
| 50 | 0.082 | 1 | 10 | .750 | .235 | 1294 | 1554 | 1814 | 2074 |
| 50 | 0.082 | 3 | 5 | .688 | .235 | 1295 | 1555 | 1815 | 2075 |
| 50 | 0.082 | 3 | 10 | .688 | .235 | 1296 | 1556 | 1816 | 2076 |
| 50 | 0.10 | 1 | 5 | .750 | .235 | 1297 | 1557 | 1817 | 2077 |
| 50 | 0.10 | 1 | 10 | .750 | .235 | 1298 | 1558 | 1818 | 2078 |
| 50 | 0.10 | 3 | 5 | .688 | .235 | 1299 | 1559 | 1819 | 2079 |
| 50 | 0.10 | 3 | 10 | .688 | .235 | 1300 | 1560 | 1820 | 2080 |
| 50 | 0.12 | 1 | 5 | .750 | .235 | 1301 | 1561 | 1821 | 2081 |
| 50 | 0.12 | 1 | 10 | .750 | .235 | 1302 | 1562 | 1822 | 2082 |
| 50 | 0.12 | 3 | 5 | .688 | .235 | 1303 | 1563 | 1823 | 2083 |
| 50 | 0.12 | 3 | 10 | .688 | .235 | 1304 | 1564 | 1824 | 2084 |
| 50 | 0.15 | 1 | 5 | .875 | .312 | 1305 | 1565 | 1825 | 2085 |
| 50 | 0.15 | 1 | 10 | .875 | .312 | 1306 | 1566 | 1826 | 2086 |
| 50 | 0.15 | 3 | 5 | .812 | .312 | 1307 | 1567 | 1827 | 2087 |
| 50 | 0.15 | 3 | 10 | .812 | .312 | 1308 | 1568 | 1828 | 2088 |
| 50 | 0.18 | 1 | 5 | .875 | .312 | 1309 | 1569 | 1829 | 2089 |
| 50 | 0.18 | 1 | 10 | .875 | .312 | 1310 | 1570 | 1830 | 2090 |
| 50 | 0.18 | 3 | 5 | .812 | .312 | 1311 | 1571 | 1831 | 2091 |
| 50 | 0.18 | 3 | 10 | .812 | .312 | 1312 | 1572 | 1832 | 2092 |
| 50 | 0.22 | 1 | 5 | .875 | .312 | 1313 | 1573 | 1833 | 2093 |
| 50 | 0.22 | 1 | 10 | .875 | .312 | 1314 | 1574 | 1834 | 2094 |
| 50 | 0.22 | 3 | 5 | .812 | .312 | 1315 | 1575 | 1835 | 2095 |
| 50 | 0.22 | 3 | 10 | .812 | .312 | 1316 | 1576 | 1836 | 2096 |

See footnotes at end of table.



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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

| DC rated voltage (Volts) | Capacitance value (μF) | Circuit diagram | Capacitance tolerance (Percent ±) | Dimension, nominal (inches) 1/ 2/ | | Dash number Failure rate level (%/1,000 hr) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | L ± 0.062 | D ± 0.031 | M(1.0) | P(0.1) | R(0.01) | S(0.001) |
| 50 | 0.27 | 1 | 5 | .875 | .312 | 1317 | 1577 | 1837 | 2097 |
| 50 | 0.27 | 1 | 10 | .875 | .312 | 1318 | 1578 | 1838 | 2098 |
| 50 | 0.27 | 3 | 5 | .812 | .312 | 1319 | 1579 | 1839 | 2099 |
| 50 | 0.27 | 3 | 10 | .812 | .312 | 1320 | 1580 | 1840 | 2100 |
| 50 | 0.33 | 1 | 5 | .875 | .312 | 1321 | 1581 | 1841 | 2101 |
| 50 | 0.33 | 1 | 10 | .875 | .312 | 1322 | 1582 | 1842 | 2102 |
| 50 | 0.33 | 3 | 5 | .812 | .312 | 1323 | 1583 | 1843 | 2103 |
| 50 | 0.33 | 3 | 10 | .812 | .312 | 1324 | 1584 | 1844 | 2104 |
| 50 | 0.39 | 1 | 5 | 1.125 | .312 | 1325 | 1585 | 1845 | 2105 |
| 50 | 0.39 | 1 | 10 | 1.125 | .312 | 1326 | 1586 | 1846 | 2106 |
| 50 | 0.39 | 3 | 5 | 1.062 | .312 | 1327 | 1587 | 1847 | 2107 |
| 50 | 0.39 | 3 | 10 | 1.062 | .312 | 1328 | 1588 | 1848 | 2108 |
| 50 | 0.47 | 1 | 5 | 1.125 | .312 | 1329 | 1589 | 1849 | 2109 |
| 50 | 0.47 | 1 | 10 | 1.125 | .312 | 1330 | 1590 | 1850 | 2110 |
| 50 | 0.47 | 3 | 5 | 1.062 | .312 | 1331 | 1591 | 1851 | 2111 |
| 50 | 0.47 | 3 | 10 | 1.062 | .312 | 1332 | 1592 | 1852 | 2112 |
| 50 | 0.56 | 1 | 5 | 1.125 | .400 | 1333 | 1593 | 1853 | 2113 |
| 50 | 0.56 | 1 | 10 | 1.125 | .400 | 1334 | 1594 | 1854 | 2114 |
| 50 | 0.56 | 3 | 5 | 1.062 | .400 | 1335 | 1595 | 1855 | 2115 |
| 50 | 0.56 | 3 | 10 | 1.062 | .400 | 1336 | 1596 | 1856 | 2116 |
| 50 | 0.68 | 1 | 5 | 1.125 | .400 | 1337 | 1597 | 1857 | 2117 |
| 50 | 0.68 | 1 | 10 | 1.125 | .400 | 1338 | 1598 | 1858 | 2118 |
| 50 | 0.68 | 3 | 5 | 1.062 | .400 | 1339 | 1599 | 1859 | 2119 |
| 50 | 0.68 | 3 | 10 | 1.062 | .400 | 1340 | 1600 | 1860 | 2120 |
| 50 | 0.82 | 1 | 5 | 1.125 | .400 | 1341 | 1601 | 1861 | 2121 |
| 50 | 0.82 | 1 | 10 | 1.125 | .400 | 1342 | 1602 | 1862 | 2122 |
| 50 | 0.82 | 3 | 5 | 1.062 | .400 | 1343 | 1603 | 1863 | 2123 |
| 50 | 0.82 | 3 | 10 | 1.062 | .400 | 1344 | 1604 | 1864 | 2124 |
| 50 | 1.0 | 1 | 5 | 1.125 | .400 | 1345 | 1605 | 1865 | 2125 |
| 50 | 1.0 | 1 | 10 | 1.125 | .400 | 1346 | 1606 | 1866 | 2126 |
| 50 | 1.0 | 3 | 5 | 1.062 | .400 | 1347 | 1607 | 1867 | 2127 |
| 50 | 1.0 | 3 | 10 | 1.062 | .400 | 1348 | 1608 | 1868 | 2128 |
| 50 | 1.2 | 1 | 5 | 1.375 | .400 | 1349 | 1609 | 1869 | 2129 |
| 50 | 1.2 | 1 | 10 | 1.375 | .400 | 1350 | 1610 | 1870 | 2130 |
| 50 | 1.2 | 3 | 5 | 1.312 | .400 | 1351 | 1611 | 1871 | 2131 |
| 50 | 1.2 | 3 | 10 | 1.312 | .400 | 1352 | 1612 | 1872 | 2132 |
| 50 | 1.5 | 1 | 5 | 1.125 | .500 | 1353 | 1613 | 1873 | 2133 |
| 50 | 1.5 | 1 | 10 | 1.125 | .500 | 1354 | 1614 | 1874 | 2134 |
| 50 | 1.5 | 3 | 5 | 1.062 | .500 | 1355 | 1615 | 1875 | 2135 |
| 50 | 1.5 | 3 | 10 | 1.062 | .500 | 1356 | 1616 | 1876 | 2136 |
| 50 | 1.8 | 1 | 5 | 1.375 | .500 | 1357 | 1617 | 1877 | 2137 |
| 50 | 1.8 | 1 | 10 | 1.375 | .500 | 1358 | 1618 | 1878 | 2138 |
| 50 | 1.8 | 3 | 5 | 1.312 | .500 | 1359 | 1619 | 1879 | 2139 |
| 50 | 1.8 | 3 | 10 | 1.312 | .500 | 1360 | 1620 | 1880 | 2140 |
| 50 | 2.2 | 1 | 5 | 1.125 | .562 | 1361 | 1621 | 1881 | 2141 |
| 50 | 2.2 | 1 | 10 | 1.125 | .562 | 1362 | 1622 | 1882 | 2142 |
| 50 | 2.2 | 3 | 5 | 1.062 | .562 | 1363 | 1623 | 1883 | 2143 |
| 50 | 2.2 | 3 | 10 | 1.062 | .562 | 1364 | 1624 | 1884 | 2144 |
| 50 | 2.7 | 1 | 5 | 1.375 | .562 | 1365 | 1625 | 1885 | 2145 |
| 50 | 2.7 | 1 | 10 | 1.375 | .562 | 1366 | 1626 | 1886 | 2146 |
| 50 | 2.7 | 3 | 5 | 1.312 | .562 | 1367 | 1627 | 1887 | 2147 |
| 50 | 2.7 | 3 | 10 | 1.312 | .562 | 1368 | 1628 | 1888 | 2148 |
| 50 | 3.3 | 1 | 5 | 1.375 | .562 | 1369 | 1629 | 1889 | 2149 |
| 50 | 3.3 | 1 | 10 | 1.375 | .562 | 1370 | 1630 | 1890 | 2150 |
| 50 | 3.3 | 3 | 5 | 1.312 | .562 | 1371 | 1631 | 1891 | 2151 |
| 50 | 3.9 | 3 | 10 | 1.312 | .562 | 1372 | 1632 | 1892 | 2152 |

See footnotes at end of table.



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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

| DC rated voltage (Volts) | Capacitance value (μF) | Circuit diagram | Capacitance tolerance (Percent ±) | Dimension, nominal (inches) 1/ 2/ | | Dash number Failure rate level (%/1,000 hr) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | L ± 0.062 | D ± 0.031 | M(1.0) | P(0.1) | R(0.01) | S(0.001) |
| 50 | 3.9 | 1 | 5 | 1.625 | .562 | 1373 | 1633 | 1893 | 2153 |
| 50 | 3.9 | 1 | 10 | 1.625 | .562 | 1374 | 1634 | 1894 | 2154 |
| 50 | 3.9 | 3 | 5 | 1.562 | .562 | 1375 | 1635 | 1895 | 2155 |
| 50 | 3.9 | 3 | 10 | 1.562 | .562 | 1376 | 1636 | 1896 | 2156 |
| 50 | 4.7 | 1 | 5 | 1.750 | .562 | 1377 | 1637 | 1897 | 2157 |
| 50 | 4.7 | 1 | 10 | 1.750 | .562 | 1378 | 1638 | 1898 | 2158 |
| 50 | 4.7 | 3 | 5 | 1.688 | .562 | 1379 | 1639 | 1899 | 2159 |
| 50 | 4.7 | 3 | 10 | 1.688 | .562 | 1380 | 1640 | 1900 | 2160 |
| 50 | 5.6 | 1 | 5 | 1.875 | .562 | 1381 | 1641 | 1901 | 2161 |
| 50 | 5.6 | 1 | 10 | 1.875 | .562 | 1382 | 1642 | 1902 | 2162 |
| 50 | 5.6 | 3 | 5 | 1.812 | .562 | 1383 | 1643 | 1903 | 2163 |
| 50 | 5.6 | 3 | 10 | 1.812 | .562 | 1384 | 1644 | 1904 | 2164 |
| 50 | 6.8 | 1 | 5 | 1.625 | .672 | 1385 | 1645 | 1905 | 2165 |
| 50 | 6.8 | 1 | 10 | 1.625 | .672 | 1386 | 1646 | 1906 | 2166 |
| 50 | 6.8 | 3 | 5 | 1.562 | .672 | 1387 | 1647 | 1907 | 2167 |
| 50 | 6.8 | 3 | 10 | 1.562 | .672 | 1388 | 1648 | 1908 | 2168 |
| 50 | 8.2 | 1 | 5 | 1.875 | .672 | 1389 | 1649 | 1909 | 2169 |
| 50 | 8.2 | 1 | 10 | 1.875 | .672 | 1390 | 1650 | 1910 | 2170 |
| 50 | 8.2 | 3 | 5 | 1.812 | .672 | 1391 | 1651 | 1911 | 2171 |
| 50 | 8.2 | 3 | 10 | 1.812 | .672 | 1392 | 1652 | 1912 | 2172 |
| 50 | 10.0 | 1 | 5 | 1.875 | .750 | 1393 | 1653 | 1913 | 2173 |
| 50 | 10.0 | 1 | 10 | 1.875 | .750 | 1394 | 1654 | 1914 | 2174 |
| 50 | 10.0 | 3 | 5 | 1.812 | .750 | 1395 | 1655 | 1915 | 2175 |
| 50 | 10.0 | 3 | 10 | 1.812 | .750 | 1396 | 1656 | 1916 | 2176 |
| 50 | 12.0 | 1 | 5 | 2.375 | .750 | 1397 | 1657 | 1917 | 2177 |
| 50 | 12.0 | 1 | 10 | 2.375 | .750 | 1398 | 1658 | 1918 | 2178 |
| 50 | 12.0 | 3 | 5 | 2.312 | .750 | 1399 | 1659 | 1919 | 2179 |
| 50 | 12.0 | 3 | 10 | 2.312 | .750 | 1400 | 1660 | 1920 | 2180 |
| 200 | 0.10 | 1 | 10 | .844 | .312 | 1401 | 1661 | 1921 | 2181 |
| 200 | 0.10 | 1 | 20 | .844 | .312 | 1402 | 1662 | 1922 | 2182 |
| 200 | 0.10 | 3 | 10 | .781 | .312 | 1403 | 1663 | 1923 | 2183 |
| 200 | 0.10 | 3 | 20 | .781 | .312 | 1404 | 1664 | 1924 | 2184 |
| 200 | 0.22 | 1 | 10 | 1.125 | .312 | 1405 | 1665 | 1925 | 2185 |
| 200 | 0.22 | 1 | 20 | 1.125 | .312 | 1406 | 1666 | 1926 | 2186 |
| 200 | 0.22 | 3 | 10 | 1.062 | .312 | 1407 | 1667 | 1927 | 2187 |
| 200 | 0.22 | 3 | 20 | 1.062 | .312 | 1408 | 1668 | 1928 | 2188 |
| 200 | 0.47 | 1 | 10 | 1.125 | .400 | 1409 | 1669 | 1929 | 2189 |
| 200 | 0.47 | 1 | 20 | 1.125 | .400 | 1410 | 1670 | 1930 | 2190 |
| 200 | 0.47 | 3 | 10 | 1.062 | .400 | 1411 | 1671 | 1931 | 2191 |
| 200 | 0.47 | 3 | 20 | 1.062 | .400 | 1412 | 1672 | 1932 | 2192 |
| 200 | 1.0 | 1 | 10 | 1.125 | .562 | 1413 | 1673 | 1933 | 2193 |
| 200 | 1.0 | 1 | 20 | 1.125 | .562 | 1414 | 1674 | 1934 | 2194 |
| 200 | 1.0 | 3 | 10 | 1.062 | .562 | 1415 | 1675 | 1935 | 2195 |
| 200 | 1.0 | 3 | 20 | 1.062 | .562 | 1416 | 1676 | 1936 | 2196 |
| 200 | 1.5 | 1 | 10 | 1.844 | .562 | 1417 | 1677 | 1937 | 2197 |
| 200 | 1.5 | 1 | 20 | 1.844 | .562 | 1418 | 1678 | 1938 | 2198 |
| 200 | 1.5 | 3 | 10 | 1.781 | .562 | 1419 | 1679 | 1939 | 2199 |
| 200 | 1.5 | 3 | 20 | 1.781 | .562 | 1420 | 1680 | 1940 | 2200 |
| 200 | 2.2 | 1 | 10 | 1.844 | .562 | 1421 | 1681 | 1941 | 2201 |
| 200 | 2.2 | 1 | 20 | 1.844 | .562 | 1422 | 1682 | 1942 | 2202 |
| 200 | 2.2 | 3 | 10 | 1.781 | .562 | 1423 | 1683 | 1943 | 2203 |
| 200 | 2.2 | 3 | 20 | 1.781 | .562 | 1424 | 1684 | 1944 | 2204 |
| 200 | 3.3 | 1 | 10 | 1.875 | .670 | 1425 | 1685 | 1945 | 2205 |
| 200 | 3.3 | 1 | 20 | 1.875 | .670 | 1426 | 1686 | 1946 | 2206 |
| 200 | 3.3 | 3 | 10 | 1.812 | .670 | 1427 | 1687 | 1947 | 2207 |
| 200 | 3.3 | 3 | 20 | 1.812 | .670 | 1428 | 1688 | 1948 | 2208 |

See footnotes at end of table.



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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

| DC rated voltage (Volts) | Capacitance value (μF) | Circuit diagram | Capacitance tolerance (Percent ±) | Dimension, nominal (inches) 1/ 2/ | | Dash number Failure rate level (%/1,000 hr) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | L ± 0.062 | D ± 0.031 | M(1.0) | P(0.1) | R(0.01) | S(0.001) |
| 200 | 4.7 | 1 | 10 | 1.875 | 1.000 | 1429 | 1689 | 1949 | 2209 |
| 200 | 4.7 | 1 | 20 | 1.875 | 1.000 | 1430 | 1690 | 1950 | 2210 |
| 200 | 4.7 | 3 | 10 | 1.812 | 1.000 | 1431 | 1691 | 1951 | 2211 |
| 200 | 4.7 | 3 | 20 | 1.812 | 1.000 | 1432 | 1692 | 1952 | 2212 |
| 200 | 6.8 | 1 | 10 | 1.875 | 1.000 | 1433 | 1693 | 1953 | 2213 |
| 200 | 6.8 | 1 | 20 | 1.875 | 1.000 | 1434 | 1694 | 1954 | 2214 |
| 200 | 6.8 | 3 | 10 | 1.812 | 1.000 | 1435 | 1695 | 1955 | 2215 |
| 200 | 6.8 | 3 | 20 | 1.812 | 1.000 | 1436 | 1696 | 1956 | 2216 |
| 200 | 10.0 | 1 | 10 | 2.375 | 1.000 | 1437 | 1697 | 1957 | 2217 |
| 200 | 10.0 | 1 | 20 | 2.375 | 1.000 | 1438 | 1698 | 1958 | 2218 |
| 200 | 10.0 | 3 | 10 | 2.312 | 1.000 | 1439 | 1699 | 1959 | 2219 |
| 200 | 10.0 | 3 | 20 | 2.312 | 1.000 | 1440 | 1700 | 1960 | 2220 |
| 200 | 12.0 | 1 | 10 | 2.625 | 1.000 | 1441 | 1701 | 1961 | 2221 |
| 200 | 12.0 | 1 | 20 | 2.625 | 1.000 | 1442 | 1702 | 1962 | 2222 |
| 200 | 12.0 | 3 | 10 | 2.562 | 1.000 | 1443 | 1703 | 1963 | 2223 |
| 200 | 12.0 | 3 | 20 | 2.562 | 1.000 | 1444 | 1704 | 1964 | 2224 |
| 400 | 0.047 | 1 | 10 | 1.125 | .312 | 1445 | 1705 | 1965 | 2225 |
| 400 | 0.047 | 1 | 20 | 1.125 | .312 | 1446 | 1706 | 1966 | 2226 |
| 400 | 0.047 | 3 | 10 | 1.062 | .312 | 1447 | 1707 | 1967 | 2227 |
| 400 | 0.047 | 3 | 20 | 1.062 | .312 | 1448 | 1708 | 1968 | 2228 |
| 400 | 0.10 | 1 | 10 | 1.125 | .400 | 1449 | 1709 | 1969 | 2229 |
| 400 | 0.10 | 1 | 20 | 1.125 | .400 | 1450 | 1710 | 1970 | 2230 |
| 400 | 0.10 | 3 | 10 | 1.062 | .400 | 1451 | 1711 | 1971 | 2231 |
| 400 | 0.10 | 3 | 20 | 1.062 | .400 | 1452 | 1712 | 1972 | 2232 |
| 400 | 0.22 | 1 | 10 | 1.125 | .562 | 1453 | 1713 | 1973 | 2233 |
| 400 | 0.22 | 1 | 20 | 1.125 | .562 | 1454 | 1714 | 1974 | 2234 |
| 400 | 0.22 | 3 | 10 | 1.062 | .562 | 1455 | 1715 | 1975 | 2235 |
| 400 | 0.22 | 3 | 20 | 1.062 | .562 | 1456 | 1716 | 1976 | 2236 |
| 400 | 0.47 | 1 | 10 | 1.625 | .562 | 1457 | 1717 | 1977 | 2237 |
| 400 | 0.47 | 1 | 20 | 1.625 | .562 | 1458 | 1718 | 1978 | 2238 |
| 400 | 0.47 | 3 | 10 | 1.562 | .562 | 1459 | 1719 | 1979 | 2239 |
| 400 | 0.47 | 3 | 20 | 1.562 | .562 | 1460 | 1720 | 1980 | 2240 |
| 400 | 1.0 | 1 | 10 | 1.875 | .750 | 1461 | 1721 | 1981 | 2241 |
| 400 | 1.0 | 1 | 20 | 1.875 | .750 | 1462 | 1722 | 1982 | 2242 |
| 400 | 1.0 | 3 | 10 | 1.812 | .750 | 1463 | 1723 | 1983 | 2243 |
| 400 | 1.0 | 3 | 20 | 1.812 | .750 | 1464 | 1724 | 1984 | 2244 |
| 400 | 2.2 | 1 | 10 | 1.875 | 1.000 | 1465 | 1725 | 1985 | 2245 |
| 400 | 2.2 | 1 | 20 | 1.875 | 1.000 | 1466 | 1726 | 1986 | 2246 |
| 400 | 2.2 | 3 | 10 | 1.812 | 1.000 | 1467 | 1727 | 1987 | 2247 |
| 400 | 2.2 | 3 | 20 | 1.812 | 1.000 | 1468 | 1728 | 1988 | 2248 |
| 400 | 3.3 | 1 | 10 | 2.625 | 1.000 | 1469 | 1729 | 1989 | 2249 |
| 400 | 3.3 | 1 | 20 | 2.625 | 1.000 | 1470 | 1730 | 1990 | 2250 |
| 400 | 3.3 | 3 | 10 | 2.625 | 1.000 | 1471 | 1731 | 1991 | 2251 |
| 400 | 3.3 | 3 | 20 | 2.625 | 1.000 | 1472 | 1732 | 1992 | 2252 |
| 400 | 3.3 | 1 | 10 | 2.625 | 1.000 | 1469 | 1729 | 1989 | 2249 |
| 400 | 3.3 | 1 | 20 | 2.625 | 1.000 | 1470 | 1730 | 1990 | 2250 |
| 400 | 3.3 | 3 | 10 | 2.562 | 1.000 | 1471 | 1731 | 1991 | 2251 |
| 400 | 3.3 | 3 | 20 | 2.562 | 1.000 | 1472 | 1732 | 1992 | 2252 |
| 600 | 0.01 | 1 | 10 | .812 | .312 | 1473 | 1733 | 1993 | 2253 |
| 600 | 0.01 | 1 | 20 | .812 | .312 | 1474 | 1734 | 1994 | 2254 |
| 600 | 0.01 | 3 | 10 | .750 | .312 | 1475 | 1735 | 1995 | 2255 |
| 600 | 0.01 | 3 | 20 | .750 | .312 | 1476 | 1736 | 1996 | 2256 |
| 600 | 0.022 | 1 | 10 | .812 | .312 | 1477 | 1737 | 1997 | 2257 |
| 600 | 0.022 | 1 | 20 | .812 | .312 | 1478 | 1738 | 1998 | 2258 |
| 600 | 0.022 | 3 | 10 | .750 | .312 | 1479 | 1739 | 1999 | 2259 |
| 600 | 0.022 | 3 | 20 | .750 | .312 | 1480 | 1740 | 2000 | 2260 |

See footnotes at end of table.



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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

| DC rated voltage (Volts) | Capacitance value (μF) | Circuit diagram | Capacitance tolerance (Percent ±) | Dimension, nominal (inches) 1/ 2/ | | Dash number Failure rate level (%/1,000 hr) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | L ± 0.062 | D ± 0.031 | M(1.0) | P(0.1) | R(0.01) | S(0.001) |
| 600 | 0.047 | 1 | 10 | 1.125 | .400 | 1481 | 1741 | 2001 | 2261 |
| 600 | 0.047 | 1 | 20 | 1.125 | .400 | 1482 | 1742 | 2002 | 2262 |
| 600 | 0.047 | 3 | 10 | 1.062 | .400 | 1483 | 1743 | 2003 | 2263 |
| 600 | 0.047 | 3 | 20 | 1.062 | .400 | 1484 | 1744 | 2004 | 2264 |
| 600 | 0.10 | 1 | 10 | 1.125 | .500 | 1485 | 1745 | 2005 | 2265 |
| 600 | 0.10 | 1 | 20 | 1.125 | .500 | 1486 | 1746 | 2006 | 2266 |
| 600 | 0.10 | 3 | 10 | 1.062 | .500 | 1487 | 1747 | 2007 | 2267 |
| 600 | 0.10 | 3 | 20 | 1.062 | .500 | 1488 | 1748 | 2008 | 2268 |
| 600 | 0.15 | 1 | 10 | 1.125 | .562 | 1509 | 1769 | 2029 | 2289 |
| 600 | 0.15 | 1 | 20 | 1.125 | .562 | 1510 | 1770 | 2030 | 2290 |
| 600 | 0.15 | 3 | 10 | 1.062 | .562 | 1511 | 1771 | 2031 | 2291 |
| 600 | 0.15 | 3 | 20 | 1.062 | .562 | 1512 | 1772 | 2032 | 2292 |
| 600 | 0.22 | 1 | 10 | 1.375 | .562 | 1489 | 1749 | 2009 | 2269 |
| 600 | 0.22 | 1 | 20 | 1.375 | .562 | 1490 | 1750 | 2010 | 2270 |
| 600 | 0.22 | 3 | 10 | 1.312 | .562 | 1491 | 1751 | 2011 | 2271 |
| 600 | 0.22 | 3 | 20 | 1.312 | .562 | 1492 | 1752 | 2012 | 2272 |
| 600 | 0.47 | 1 | 10 | 1.625 | .670 | 1493 | 1753 | 2013 | 2273 |
| 600 | 0.47 | 1 | 20 | 1.625 | .670 | 1494 | 1754 | 2014 | 2274 |
| 600 | 0.47 | 3 | 10 | 1.562 | .670 | 1495 | 1755 | 2015 | 2275 |
| 600 | 0.47 | 3 | 20 | 1.562 | .670 | 1496 | 1756 | 2016 | 2276 |
| 600 | 1.0 | 1 | 10 | 1.844 | 1.000 | 1497 | 1757 | 2017 | 2277 |
| 600 | 1.0 | 1 | 20 | 1.844 | 1.000 | 1498 | 1758 | 2018 | 2278 |
| 600 | 1.0 | 3 | 10 | 1.781 | 1.000 | 1499 | 1759 | 2019 | 2279 |
| 600 | 1.0 | 3 | 20 | 1.781 | 1.000 | 1500 | 1760 | 2020 | 2280 |
| 600 | 1.5 | 1 | 10 | 1.875 | 1.000 | 1501 | 1761 | 2021 | 2281 |
| 600 | 1.5 | 1 | 20 | 1.875 | 1.000 | 1502 | 1762 | 2022 | 2282 |
| 600 | 1.5 | 3 | 10 | 1.812 | 1.000 | 1503 | 1763 | 2023 | 2283 |
| 600 | 1.5 | 3 | 20 | 1.812 | 1.000 | 1504 | 1764 | 2024 | 2284 |
| 600 | 2.2 | 1 | 10 | 2.625 | 1.000 | 1505 | 1765 | 2025 | 2285 |
| 600 | 2.2 | 1 | 20 | 2.625 | 1.000 | 1506 | 1766 | 2026 | 2286 |
| 600 | 2.2 | 3 | 10 | 2.562 | 1.000 | 1507 | 1767 | 2027 | 2287 |
| 600 | 2.2 | 3 | 20 | 2.562 | 1.000 | 1508 | 1768 | 2028 | 2288 |

1/ Dimensions shown are bare case sizes (see figure 1).
2/ Metric equivalents, see table III.

TABLE II. Characteristics.

| Dielectric material | Paper-polyethylene-terephthalate | | Polyethylene-terephthalate |
|---|---|---|---|
| Operating temperature range | -55°C to +125°C | | -55°C to +85°C |
| DC voltage rating (volts) | 200 V | 400 V, 600 V | 50 V |
| High ambient test temperature<br>Low ambient test temperature | +125°C +4°C, -0°C<br>-55°C +0°C, -3°C | +125°C +4°C, -0°C<br>-55°C +0°C, -3°C | +85°C +4°C, -0°C<br>-55°C +0°C, -3°C |
| Megohms x microfarads (minimum)<br>At +25°C<br>At high ambient test temperature | <br>2,000<br>10 | <br>2,000<br>40 | <br>25,000<br>3,000 |
| Insulation resistance (megohms) (need not exceed)<br>At +25°C<br>At high ambient test temperature | <br>12,000<br>150 | <br>12,000<br>600 | <br>50,000<br>6,000 |
| Capacitance change with temperature (maximum)<br>At -55°C<br>At high ambient test temperature | <br>-10<br>+20 | <br>-10<br>+20 | <br>-8 percent<br>+8 percent |



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TABLE III. Metric equivalents of decimal inches. 1/

| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .031 | 0.79 | .750 | 19.05 | 1.688 | 42.88 |
| .062 | 1.57 | .781 | 19.84 | 1.750 | 44.45 |
| .175 | 4.45 | .812 | 20.62 | 1.781 | 45.24 |
| .195 | 4.95 | .844 | 21.44 | 1.812 | 46.02 |
| .235 | 5.97 | .875 | 22.23 | 1.844 | 46.84 |
| .312 | 7.92 | 1.000 | 25.40 | 1.875 | 47.63 |
| .400 | 10.16 | 1.062 | 26.97 | 2.312 | 58.72 |
| .500 | 12.70 | 1.125 | 28.58 | 2.375 | 60.33 |
| .562 | 14.27 | 1.312 | 33.32 | 2.562 | 65.07 |
| .670 | 17.02 | 1.375 | 34.93 | 2.625 | 66.68 |
| .672 | 17.07 | 1.562 | 39.67 | | |
| .688 | 17.48 | 1.625 | 41.28 | | |

1/ Metric equivalents are given for general information only.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 feet).

125 percent of dc rated voltage applied: See MIL-PRF-39022 for voltage limitations.

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 g's peak).

50 percent of rated voltage applied.

Salt atmosphere (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Salt solution: 5 percent.

Immersion: Method 104 of MIL-STD-202, condition C.

DWV:

Insulating sleeves: Greater than 4,000 volts, dc.

Terminal to terminal: 200 percent of rated voltage.

Terminals to case (when case is not a terminal): 200 percent of rated voltage.

IR:

Insulating sleeves: 100 megohms, minimum.

Terminal to terminal: Not less than 50 percent of value specified in table II.

Terminals to case (when case is not a terminal): 5,000 megohms, minimum.

Capacitance: Within ± 10 percent of initial value.

DF: Not more than 1.1 percent.

Solderability: Method 208 of MIL-STD-202.



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Shock (specified pulse): Method 213 of MIL-STD-202, condition I.

50 percent of rated voltage applied.

Moisture resistance: Method 106 of MIL-STD-202.

DWV, IR, Capacitance, and DF: Same as for immersion.

Terminal strength: Method 211 of MIL-STD-202, conditions A and D.

Condition A: Applied force of 5 pounds.

Condition D: 3 rotations of 360 degrees.

Low temperature and capacitance change with temperature:

Low temperature: -55°C +0°C, -3°C for 48 hours ± 4 hours with rated voltage applied.

Capacitance change with temperature:

-55°C +0°C, -3°C: See table II.

At applicable high test temperature: See table II.

Fungus: Method 508 of MIL-STD-810.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition C, except the time shall be 10 seconds ± 1 second.

IR: See table II.

Capacitance: Within ± 5 percent of initial value.

DF: Not more than 1.0 percent.

Life: Method 108 of MIL-STD-202.

Qualification:

Accelerated conditions: 140 percent of rated voltage for 2,000 +72, -0 hours.

Rated conditions: 100 percent of rated voltage for 10,000 +96, -0 hours.

DF (at applicable high test temperature) between 24 and 48 hours of test: 2 percent, maximum.

During last 48 hours of test: 2.5 percent, maximum.

IR:

Insulating sleeves: 100 megohms, minimum.

Terminal to terminal: Not less than 33.3 percent of initial requirement.

Terminals to case (when case is not a terminal): 5,000 megohms, minimum.

Capacitance: Within ± 10 percent of initial value.

DF (at +25°C ± 3°C) after life: 1.25 percent.



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Quality conformance: 240 hours at accelerated conditions and meet the requirements for qualification test.

Extended life:

Accelerated conditions: 2,000 +72, -0 hours.

Rated conditions: 10,000 +96, -0 hours.

Substitutability: See table IV.

Marking: In accordance with MIL-PRF-39022.

Part or Identifying Number (PIN): M39022/01- (dash number from table I).

TABLE IV. Substitutability data.

| Dash number in MIL-C-39022/1C thru MIL-PRF-39022/1G 1/ | | | | | Type designation in MIL-C-39022/1B 2/ | Dash number in MIL-C-39022/1A and /1 |
|---|---|---|---|---|---|---|
| L (5.0) | M (1.0) | P (0.1) | R (0.01) | S (0.001) | | |
| 1001 | 1261 | 1521 | 1781 | 2041 | CHR09A1RA183-- | 0001 |
| 1002 | 1262 | 1522 | 1782 | 2042 | CHR09A1RA183-- | 0002 |
| 1003 | 1263 | 1523 | 1783 | 2043 | CHR09A3RA183-- | 0003 |
| 1004 | 1264 | 1524 | 1784 | 2044 | CHR09A3RA183-- | 0004 |
| 1005 | 1265 | 1525 | 1785 | 2045 | CHR09A1RA223-- | 0005 |
| 1006 | 1266 | 1526 | 1786 | 2046 | CHR09A1RA223-- | 0006 |
| 1007 | 1267 | 1527 | 1787 | 2047 | CHR09A3RA223-- | 0007 |
| 1008 | 1268 | 1528 | 1788 | 2048 | CHR09A3RA223-- | 0008 |
| 1009 | 1269 | 1529 | 1789 | 2049 | CHR09A1RA273-- | 0009 |
| 1010 | 1270 | 1530 | 1790 | 2050 | CHR09A1RA273-- | 0010 |
| 1011 | 1271 | 1531 | 1791 | 2051 | CHR09A3RA273-- | 0011 |
| 1012 | 1272 | 1532 | 1792 | 2052 | CHR09A3RA273-- | 0012 |
| 1013 | 1273 | 1533 | 1793 | 2053 | CHR09A1RA333-- | 0013 |
| 1014 | 1274 | 1534 | 1794 | 2054 | CHR09A1RA333-- | 0014 |
| 1015 | 1275 | 1535 | 1795 | 2055 | CHR09A3RA333-- | 0015 |
| 1016 | 1276 | 1536 | 1796 | 2056 | CHR09A3RA333-- | 0016 |
| 1017 | 1277 | 1537 | 1797 | 2057 | CHR09A1RA393-- | 0017 |
| 1018 | 1278 | 1538 | 1798 | 2058 | CHR09A1RA393-- | 0018 |
| 1019 | 1279 | 1539 | 1799 | 2059 | CHR09A3RA393-- | 0019 |
| 1020 | 1280 | 1540 | 1800 | 2060 | CHR09A3RA393-- | 0020 |
| 1021 | 1281 | 1541 | 1801 | 2061 | CHR09A1RA473-- | 0021 |
| 1022 | 1282 | 1542 | 1802 | 2062 | CHR09A1RA473-- | 0022 |
| 1023 | 1283 | 1543 | 1803 | 2063 | CHR09A3RA473-- | 0023 |
| 1024 | 1284 | 1544 | 1804 | 2064 | CHR09A3RA473-- | 0024 |
| 1025 | 1285 | 1545 | 1805 | 2065 | CHR09A1RA563-- | 0025 |
| 1026 | 1286 | 1546 | 1806 | 2066 | CHR09A1RA563-- | 0026 |
| 1027 | 1287 | 1547 | 1807 | 2067 | CHR09A3RA563-- | 0027 |
| 1028 | 1288 | 1548 | 1808 | 2068 | CHR09A3RA563-- | 0028 |
| 1029 | 1289 | 1549 | 1809 | 2069 | CHR09A1RA683-- | 0029 |
| 1030 | 1290 | 1550 | 1810 | 2070 | CHR09A1RA683-- | 0030 |
| 1031 | 1291 | 1551 | 1811 | 2071 | CHR09A3RA683-- | 0031 |
| 1032 | 1292 | 1552 | 1812 | 2072 | CHR09A3RA683-- | 0032 |
| 1033 | 1293 | 1553 | 1813 | 2073 | CHR09A1RA823-- | 0033 |
| 1034 | 1294 | 1554 | 1814 | 2074 | CHR09A1RA823-- | 0034 |

See footnote at end of table.



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TABLE IV. Substitutability data - Continued.

| Dash number in MIL-C-39022/1C thru MIL-PRF-39022/1G 1/ | | | | | Type designation in MIL-C-39022/1B 2/ | Dash number in MIL-C-39022/1A and /1 |
|---|---|---|---|---|---|---|
| L (5.0) | M (1.0) | P (0.1) | R (0.01) | S (0.001) | | |
| 1035 | 1295 | 1555 | 1815 | 2075 | CHR09A3RA823-- | 0035 |
| 1036 | 1296 | 1556 | 1816 | 2076 | CHR09A3RA823-- | 0036 |
| 1037 | 1297 | 1557 | 1817 | 2077 | CHR09A1RA104-- | 0037 |
| 1038 | 1298 | 1558 | 1818 | 2078 | CHR09A1RA104-- | 0038 |
| 1039 | 1299 | 1559 | 1819 | 2079 | CHR09A3RA104-- | 0039 |
| 1040 | 1300 | 1560 | 1820 | 2080 | CHR09A3RA104-- | 0040 |
| 1041 | 1301 | 1561 | 1821 | 2081 | CHR09A1RA124-- | |
| 1042 | 1302 | 1562 | 1822 | 2082 | CHR09A1RA124-- | |
| 1043 | 1303 | 1563 | 1823 | 2083 | CHR09A3RA124-- | |
| 1044 | 1304 | 1564 | 1824 | 2084 | CHR09A3RA124-- | |
| 1045 | 1305 | 1565 | 1825 | 2085 | CHR09A1RA154-- | |
| 1046 | 1306 | 1566 | 1826 | 2086 | CHR09A1RA154-- | |
| 1047 | 1307 | 1567 | 1827 | 2087 | CHR09A3RA154-- | |
| 1048 | 1308 | 1568 | 1828 | 2088 | CHR09A3RA154-- | |
| 1049 | 1309 | 1569 | 1829 | 2089 | CHR09A1RA184-- | 0041 |
| 1050 | 1310 | 1570 | 1830 | 2090 | CHR09A1RA184-- | 0042 |
| 1051 | 1311 | 1571 | 1831 | 2091 | CHR09A3RA184-- | 0043 |
| 1052 | 1312 | 1572 | 1832 | 2092 | CHR09A3RA184-- | 0044 |
| 1053 | 1313 | 1573 | 1833 | 2093 | CHR09A1RA224-- | 0045 |
| 1054 | 1314 | 1574 | 1834 | 2094 | CHR09A1RA224-- | 0046 |
| 1055 | 1315 | 1575 | 1835 | 2095 | CHR09A3RA224-- | 0047 |
| 1056 | 1316 | 1576 | 1836 | 2096 | CHR09A3RA224-- | 0048 |
| 1057 | 1317 | 1577 | 1837 | 2097 | CHR09A1RA274-- | 0049 |
| 1058 | 1318 | 1578 | 1838 | 2098 | CHR09A1RA274-- | 0050 |
| 1059 | 1319 | 1579 | 1839 | 2099 | CHR09A3RA274-- | 0051 |
| 1060 | 1320 | 1580 | 1840 | 2100 | CHR09A3RA274-- | 0052 |
| 1061 | 1321 | 1581 | 1841 | 2101 | CHR09A1RA334-- | 0053 |
| 1062 | 1322 | 1582 | 1842 | 2102 | CHR09A1RA334-- | 0054 |
| 1063 | 1323 | 1583 | 1843 | 2103 | CHR09A3RA334-- | 0055 |
| 1064 | 1324 | 1584 | 1844 | 2104 | CHR09A3RA334-- | 0056 |
| 1065 | 1325 | 1585 | 1845 | 2105 | CHR09A1RA394-- | 0057 |
| 1066 | 1326 | 1586 | 1846 | 2106 | CHR09A1RA394-- | 0058 |
| 1067 | 1327 | 1587 | 1847 | 2107 | CHR09A3RA394-- | 0059 |
| 1068 | 1328 | 1588 | 1848 | 2108 | CHR09A3RA394-- | 0060 |
| 1069 | 1329 | 1589 | 1849 | 2109 | CHR09A1RA474-- | 0061 |
| 1070 | 1330 | 1590 | 1850 | 2110 | CHR09A1RA474-- | 0062 |
| 1071 | 1331 | 1591 | 1851 | 2111 | CHR09A3RA474-- | 0063 |
| 1072 | 1332 | 1592 | 1852 | 2112 | CHR09A3RA474-- | 0064 |
| 1073 | 1333 | 1593 | 1853 | 2113 | CHR09A1RA564-- | 0065 |
| 1074 | 1334 | 1594 | 1854 | 2114 | CHR09A1RA564-- | 0066 |
| 1075 | 1335 | 1595 | 1855 | 2115 | CHR09A3RA564-- | 0067 |
| 1076 | 1336 | 1596 | 1856 | 2116 | CHR09A3RA564-- | 0068 |
| 1077 | 1337 | 1597 | 1857 | 2117 | CHR09A1RA684-- | 0069 |
| 1078 | 1338 | 1598 | 1858 | 2118 | CHR09A1RA684-- | 0070 |
| 1079 | 1339 | 1599 | 1859 | 2119 | CHR09A3RA684-- | 0071 |
| 1080 | 1340 | 1600 | 1860 | 2120 | CHR09A3RA684-- | 0072 |

See footnote at end of table.



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TABLE IV. Substitutability data - Continued.

| Dash number in MIL-C-39022/1C thru MIL-PRF-39022/1G 1/ | | | | | Type designation in MIL-C-39022/1B 2/ | Dash number in MIL-C-39022/1A and /1 |
|---|---|---|---|---|---|---|
| L (5.0) | M (1.0) | P (0.1) | R (0.01) | S (0.001) | | |
| 1081 | 1341 | 1601 | 1861 | 2121 | CHR09A1RA824-- | 0073 |
| 1082 | 1342 | 1602 | 1862 | 2122 | CHR09A1RA824-- | 0074 |
| 1083 | 1343 | 1603 | 1863 | 2123 | CHR09A3RA824-- | 0075 |
| 1084 | 1344 | 1604 | 1864 | 2124 | CHR09A3RA824-- | 0076 |
| 1085 | 1345 | 1605 | 1865 | 2125 | CHR09A1RA105-- | 0077 |
| 1086 | 1346 | 1606 | 1866 | 2126 | CHR09A1RA105-- | 0078 |
| 1087 | 1347 | 1607 | 1867 | 2127 | CHR09A3RA105-- | 0079 |
| 1088 | 1348 | 1608 | 1868 | 2128 | CHR09A3RA105-- | 0080 |
| 1089 | 1349 | 1609 | 1869 | 2129 | CHR09A1RA125-- | |
| 1090 | 1350 | 1610 | 1870 | 2130 | CHR09A1RA125-- | |
| 1091 | 1351 | 1611 | 1871 | 2131 | CHR09A3RA125-- | |
| 1092 | 1352 | 1612 | 1872 | 2132 | CHR09A3RA125-- | |
| 1093 | 1353 | 1613 | 1873 | 2133 | CHR09A1RA155-- | 0081 |
| 1094 | 1354 | 1614 | 1874 | 2134 | CHR09A1RA155-- | 0082 |
| 1095 | 1355 | 1615 | 1875 | 2135 | CHR09A3RA155-- | 0083 |
| 1096 | 1356 | 1616 | 1876 | 2136 | CHR09A3RA155-- | 0084 |
| 1097 | 1357 | 1617 | 1877 | 2137 | CHR09A1RA185-- | |
| 1098 | 1358 | 1618 | 1878 | 2138 | CHR09A1RA185-- | |
| 1099 | 1359 | 1619 | 1879 | 2139 | CHR09A3RA185-- | |
| 1100 | 1360 | 1620 | 1880 | 2140 | CHR09A3RA185-- | |
| 1101 | 1361 | 1621 | 1881 | 2141 | CHR09A1RA225-- | |
| 1102 | 1362 | 1622 | 1882 | 2142 | CHR09A1RA225-- | |
| 1103 | 1363 | 1623 | 1883 | 2143 | CHR09A3RA225-- | |
| 1104 | 1364 | 1624 | 1884 | 2144 | CHR09A3RA225-- | |
| 1105 | 1365 | 1625 | 1885 | 2145 | CHR09A1RA275-- | |
| 1106 | 1366 | 1626 | 1886 | 2146 | CHR09A1RA275-- | |
| 1107 | 1367 | 1627 | 1887 | 2147 | CHR09A3RA275-- | |
| 1108 | 1368 | 1628 | 1888 | 2148 | CHR09A3RA275-- | |
| 1109 | 1369 | 1629 | 1889 | 2149 | CHR09A1RA335-- | |
| 1110 | 1370 | 1630 | 1890 | 2150 | CHR09A1RA335-- | |
| 1111 | 1371 | 1631 | 1891 | 2151 | CHR09A3RA335-- | |
| 1112 | 1372 | 1632 | 1892 | 2152 | CHR09A3RA335-- | |
| 1113 | 1373 | 1633 | 1893 | 2153 | CHR09A1RA395-- | |
| 1114 | 1374 | 1634 | 1894 | 2154 | CHR09A1RA395-- | |
| 1115 | 1375 | 1635 | 1895 | 2155 | CHR09A3RA395-- | |
| 1116 | 1376 | 1636 | 1896 | 2156 | CHR09A3RA395-- | |
| 1117 | 1377 | 1637 | 1897 | 2157 | CHR09A1RA475-- | |
| 1118 | 1378 | 1638 | 1898 | 2158 | CHR09A1RA475-- | |
| 1119 | 1379 | 1639 | 1899 | 2159 | CHR09A3RA475-- | |
| 1120 | 1380 | 1640 | 1900 | 2160 | CHR09A3RA475-- | |
| 1121 | 1381 | 1641 | 1901 | 2161 | CHR09A1RA565-- | |
| 1122 | 1382 | 1642 | 1902 | 2162 | CHR09A1RA565-- | |
| 1123 | 1383 | 1643 | 1903 | 2163 | CHR09A3RA565-- | |
| 1124 | 1384 | 1644 | 1904 | 2164 | CHR09A3RA565-- | |
| 1125 | 1385 | 1645 | 1905 | 2165 | CHR09A1RA685-- | |
| 1126 | 1386 | 1646 | 1906 | 2166 | CHR09A1RA685-- | |

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TABLE IV. Substitutability data - Continued.

| Dash number in MIL-C-39022/1C thru MIL-PRF-39022/1G 1/ | | | | | Type designation in MIL-C-39022/1B 2/ | Dash number in MIL-C-39022/1A and /1 |
|---|---|---|---|---|---|---|
| L (5.0) | M (1.0) | P (0.1) | R (0.01) | S (0.001) | | |
| 1127 | 1387 | 1647 | 1907 | 2167 | CHR09A3RA685-- | |
| 1128 | 1388 | 1648 | 1908 | 2168 | CHR09A3RA685-- | |
| 1129 | 1389 | 1649 | 1909 | 2169 | CHR09A1RA825-- | |
| 1130 | 1390 | 1650 | 1910 | 2170 | CHR09A1RA825-- | |
| 1131 | 1391 | 1651 | 1911 | 2171 | CHR09A3RA825-- | |
| 1132 | 1392 | 1652 | 1912 | 2172 | CHR09A3RA825-- | |
| 1133 | 1393 | 1653 | 1913 | 2173 | CHR09A1RA106-- | 0113 |
| 1134 | 1394 | 1654 | 1914 | 2174 | CHR09A1RA106-- | 0114 |
| 1135 | 1395 | 1655 | 1915 | 2175 | CHR09A3RA106-- | 0115 |
| 1136 | 1396 | 1656 | 1916 | 2176 | CHR09A3RA106-- | 0116 |
| 1137 | 1397 | 1657 | 1917 | 2177 | CHR09A1RA126-- | 0117 |
| 1138 | 1398 | 1658 | 1918 | 2178 | CHR09A1RA126-- | 0118 |
| 1139 | 1399 | 1659 | 1919 | 2179 | CHR09A3RA126-- | 0119 |
| 1140 | 1400 | 1660 | 1920 | 2180 | CHR09A3RA126-- | 0120 |
| 1141 | 1401 | 1661 | 1921 | 2181 | CHR09A1NC104-- | |
| 1142 | 1402 | 1662 | 1922 | 2182 | CHR09A1NC104-- | |
| 1143 | 1403 | 1663 | 1923 | 2183 | CHR09A3NC104-- | |
| 1144 | 1404 | 1664 | 1924 | 2184 | CHR09A3NC104-- | |
| 1145 | 1405 | 1665 | 1925 | 2185 | CHR09A1NC224-- | |
| 1146 | 1406 | 1666 | 1926 | 2186 | CHR09A1NC224-- | |
| 1147 | 1407 | 1667 | 1927 | 2187 | CHR09A3NC224-- | |
| 1148 | 1408 | 1668 | 1928 | 2188 | CHR09A3NC224-- | |
| 1149 | 1409 | 1669 | 1929 | 2189 | CHR09A1NC474-- | |
| 1150 | 1410 | 1670 | 1930 | 2190 | CHR09A1NC474-- | |
| 1157 | 1417 | 1677 | 1937 | 2197 | CHR09A1NC155-- | |
| 1158 | 1418 | 1678 | 1938 | 2198 | CHR09A1NC155-- | |
| 1159 | 1419 | 1679 | 1939 | 2199 | CHR09A3NC155-- | |
| 1160 | 1420 | 1680 | 1940 | 2200 | CHR09A3NC155-- | |
| 1161 | 1421 | 1681 | 1941 | 2201 | CHR09A1NC225-- | |
| 1162 | 1422 | 1682 | 1942 | 2202 | CHR09A1NC225-- | |
| 1163 | 1423 | 1683 | 1943 | 2203 | CHR09A3NC225-- | |
| 1164 | 1424 | 1684 | 1944 | 2204 | CHR09A3NC225-- | |
| 1165 | 1425 | 1685 | 1945 | 2205 | CHR09A1NC335-- | |
| 1166 | 1426 | 1686 | 1946 | 2206 | CHR09A1NC335-- | |
| 1167 | 1427 | 1687 | 1947 | 2207 | CHR09A3NC335-- | |
| 1168 | 1428 | 1688 | 1948 | 2208 | CHR09A3NC335-- | |
| 1169 | 1429 | 1689 | 1949 | 2209 | CHR09A1NC475-- | |
| 1170 | 1430 | 1690 | 1950 | 2210 | CHR09A1NC475-- | |
| 1171 | 1431 | 1691 | 1951 | 2211 | CHR09A3NC475-- | |
| 1172 | 1432 | 1692 | 1952 | 2212 | CHR09A3NC475-- | |
| 1173 | 1433 | 1693 | 1953 | 2213 | CHR09A1NC685-- | |
| 1174 | 1434 | 1694 | 1954 | 2214 | CHR09A1NC685-- | |
| 1175 | 1435 | 1695 | 1955 | 2215 | CHR09A3NC685-- | |
| 1176 | 1436 | 1696 | 1956 | 2216 | CHR09A3NC685-- | |
| 1177 | 1437 | 1697 | 1957 | 2217 | CHR09A1NC106-- | |
| 1178 | 1438 | 1698 | 1958 | 2218 | CHR09A1NC106-- | |

See footnote at end of table.



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TABLE IV. Substitutability data - Continued.

| Dash number in MIL-C-39022/1C thru MIL-PRF-39022/1G 1/ | | | | | Type designation in MIL-C-39022/1B 2/ | Dash number in MIL-C-39022/1A and /1 |
|---|---|---|---|---|---|---|
| L (5.0) | M (1.0) | P (0.1) | R (0.01) | S (0.001) | | |
| 1179 | 1439 | 1699 | 1959 | 2219 | CHR09A3NC106-- | |
| 1180 | 1440 | 1700 | 1960 | 2220 | CHR09A3NC106-- | |
| 1181 | 1441 | 1701 | 1961 | 2221 | CHR09A1NC126-- | |
| 1182 | 1442 | 1702 | 1962 | 2222 | CHR09A1NC126-- | |
| 1183 | 1443 | 1703 | 1963 | 2223 | CHR09A3NC126-- | |
| 1184 | 1444 | 1704 | 1964 | 2224 | CHR09A3NC126-- | |
| 1185 | 1445 | 1705 | 1965 | 2225 | CHR09A1NE473-- | |
| 1186 | 1446 | 1706 | 1966 | 2226 | CHR09A1NE473-- | |
| 1187 | 1447 | 1707 | 1967 | 2227 | CHR09A3NE473-- | |
| 1188 | 1448 | 1708 | 1968 | 2228 | CHR09A3NE473-- | |
| 1189 | 1449 | 1709 | 1969 | 2229 | CHR09A1NE104-- | |
| 1190 | 1450 | 1710 | 1970 | 2230 | CHR09A1NE104-- | |
| 1191 | 1451 | 1711 | 1971 | 2231 | CHR09A3NE104-- | |
| 1192 | 1452 | 1712 | 1972 | 2232 | CHR09A3NE104-- | |
| 1193 | 1453 | 1713 | 1973 | 2233 | CHR09A1NE224-- | |
| 1194 | 1454 | 1714 | 1974 | 2234 | CHR09A1NE224-- | |
| 1195 | 1455 | 1715 | 1975 | 2235 | CHR09A3NE224-- | |
| 1196 | 1456 | 1716 | 1976 | 2236 | CHR09A3NE224-- | |
| 1197 | 1457 | 1717 | 1977 | 2237 | CHR09A1NE474-- | |
| 1198 | 1458 | 1718 | 1978 | 2238 | CHR09A1NE474-- | |
| 1199 | 1459 | 1719 | 1979 | 2239 | CHR09A3NE474-- | |
| 1200 | 1460 | 1720 | 1980 | 2240 | CHR09A3NE474-- | |
| 1201 | 1461 | 1721 | 1981 | 2241 | CHR09A1NE105-- | |
| 1202 | 1462 | 1722 | 1982 | 2242 | CHR09A1NE105-- | |
| 1203 | 1463 | 1723 | 1983 | 2243 | CHR09A3NE105-- | |
| 1204 | 1464 | 1724 | 1984 | 2244 | CHR09A3NE105-- | |
| 1205 | 1465 | 1725 | 1985 | 2245 | CHR09A1NE125-- | |
| 1206 | 1466 | 1726 | 1986 | 2246 | CHR09A1NE125-- | |
| 1207 | 1467 | 1727 | 1987 | 2247 | CHR09A3NE125-- | |
| 1208 | 1468 | 1728 | 1988 | 2248 | CHR09A3NE125-- | |
| 1209 | 1469 | 1729 | 1989 | 2249 | CHR09A1NE335-- | |
| 1210 | 1470 | 1730 | 1990 | 2250 | CHR09A1NE335-- | |
| 1211 | 1471 | 1731 | 1991 | 2251 | CHR09A3NE335-- | |
| 1212 | 1472 | 1732 | 1992 | 2252 | CHR09A3NE335-- | |
| 1213 | 1473 | 1733 | 1993 | 2253 | CHR09A1NF103-- | |
| 1214 | 1474 | 1734 | 1994 | 2254 | CHR09A1NF103-- | |
| 1215 | 1475 | 1735 | 1995 | 2255 | CHR09A3NF103-- | |
| 1216 | 1476 | 1736 | 1996 | 2256 | CHR09A3NF103-- | |
| 1217 | 1477 | 1737 | 1997 | 2257 | CHR09A1NF223-- | |
| 1218 | 1478 | 1738 | 1998 | 2258 | CHR09A1NF223-- | |
| 1219 | 1479 | 1739 | 1999 | 2259 | CHR09A3NF223-- | |
| 1220 | 1480 | 1740 | 2000 | 2260 | CHR09A3NF223-- | |
| 1221 | 1481 | 1741 | 2001 | 2261 | CHR09A1NF473-- | |
| 1222 | 1482 | 1742 | 2002 | 2262 | CHR09A1NF473-- | |
| 1223 | 1483 | 1743 | 2003 | 2263 | CHR09A3NF473-- | |
| 1224 | 1484 | 1744 | 2004 | 2264 | CHR09A3NF473-- | |

See footnote at end of table.



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TABLE IV. Substitutability data - Continued.

| Dash number in MIL-C-39022/1C thru MIL-PRF-39022/1G 1/ | | | | | Type designation in MIL-C-39022/1B 2/ | Dash number in MIL-C-39022/1A and /1 |
|---|---|---|---|---|---|---|
| L (5.0) | M (1.0) | P (0.1) | R (0.01) | S (0.001) | | |
| 1225 | 1485 | 1745 | 2005 | 2265 | CHR09A1NF104-- | |
| 1226 | 1486 | 1746 | 2006 | 2266 | CHR09A1NF104-- | |
| 1227 | 1487 | 1747 | 2007 | 2267 | CHR09A3NF104-- | |
| 1228 | 1488 | 1748 | 2008 | 2268 | CHR09A3NF104-- | |
| 1249 | 1509 | 1769 | 2029 | 2289 | | |
| 1250 | 1510 | 1770 | 2030 | 2290 | | |
| 1251 | 1511 | 1771 | 2031 | 2291 | | |
| 1252 | 1512 | 1772 | 2032 | 2292 | | |
| 1229 | 1489 | 1749 | 2009 | 2269 | CHR09A1NF224-- | |
| 1231 | 1490 | 1750 | 2010 | 2270 | CHR09A1NF224-- | |
| 1230 | 1491 | 1751 | 2011 | 2271 | CHR09A3NF224-- | |
| 1232 | 1492 | 1752 | 2012 | 2272 | CHR09A3NF224-- | |
| 1233 | 1493 | 1753 | 2013 | 2273 | CHR09A1NF474-- | |
| 1234 | 1494 | 1754 | 2014 | 2274 | CHR09A1NF474-- | |
| 1235 | 1495 | 1755 | 2015 | 2275 | CHR09A3NF474-- | |
| 1236 | 1496 | 1756 | 2016 | 2276 | CHR09A3NF474-- | |
| 1237 | 1497 | 1757 | 2017 | 2277 | CHR09A1NF105-- | |
| 1238 | 1498 | 1758 | 2018 | 2278 | CHR09A1NF105-- | |
| 1239 | 1499 | 1759 | 2019 | 2279 | CHR09A3NF105-- | |
| 1240 | 1500 | 1760 | 2020 | 2280 | CHR09A3NF105-- | |
| 1241 | 1501 | 1761 | 2021 | 2281 | CHR09A1NF155-- | |
| 1242 | 1502 | 1762 | 2022 | 2282 | CHR09A1NF155-- | |
| 1243 | 1503 | 1763 | 2023 | 2283 | CHR09A3NF155-- | |
| 1244 | 1504 | 1764 | 2024 | 2284 | CHR09A3NF155-- | |
| 1245 | 1505 | 1765 | 2025 | 2285 | CHR09A1NF225-- | |
| 1246 | 1506 | 1766 | 2026 | 2286 | CHR09A1NF225-- | |
| 1247 | 1507 | 1767 | 2027 | 2287 | CHR09A3NF225-- | |
| 1248 | 1508 | 1768 | 2028 | 2288 | CHR09A3NF225-- | |

1/ FRL L for revision C only.
2/ Complete type designation will include additional symbols for capacitance tolerance and failure rate level.

Changes form previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents: In addition to MIL-PRF-39022, this document also references MIL-STD-202 and MIL-STD-810.



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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5910-2011-034)

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